

Typeface–Context Congruence and the Cohort-Dependent Basis of Font Credibility

Abstract

Readers judge the same words as more or less credible depending on the typeface that carries them, but it is unclear whether such judgments reflect fixed connotations of individual typefaces or the fit between a typeface and the kind of message it renders. In a preregistered online experiment, adults from two age cohorts rated short institutional and interpersonal messages displayed in either a serif or a sans-serif typeface, drawn from several typeface families. Perceived credibility depended less on the typeface itself than on whether the typeface matched the message's communicative register, and this sensitivity to typographic fit was more pronounced in older than in younger adults. Secondary measures helped distinguish credibility judgments from related evaluations of typographic fit and readability. The results caution against treating any typeface as intrinsically credible and imply that effective typography for mixed-age audiences depends on message context rather than on typeface alone.

Significance. Practitioners often follow a simple heuristic—serif for authority, sans-serif for modernity—as though credibility were a stable property of a typeface. We find instead that a typeface earns or loses credibility largely through its fit with the communicative situation, and that audiences differ in how heavily a mismatch is penalized. The findings suggest that typography guidelines should consider both message context and audience composition rather than assigning fixed psychological properties to font categories.

Introduction

A lease clause set in a serif face and the same clause set in a sans-serif face carry identical legal content, yet readers do not respond to them identically. Typefaces convey connotative meaning that travels independently of the words they spell: they are perceived as having distinct “personalities” (Brumberger, 2003; Shaikh et al., 2006), they color the emotional and persuasive reading of a text (Juni & Gross, 2008), and they shift brand and product judgments (Childers & Jass, 2002). More broadly, surface features of a message, not only its content, drive how credible it seems (Fogg et al., 2003), where credibility blends perceptions of competence and trustworthiness (Fiske et al., 2007).

How this connotation bears on credibility admits two accounts. On an **intrinsic-cue** account, a typeface (or a broad class such as serif) carries a stable connotation—serif as authoritative, say—so serif faces should read as more credible wherever they appear. On a **congruence** account, connotation matters only relative to the message: credibility rises when a typeface's connotations fit the message's communicative register and falls when they clash (Childers & Jass, 2002; Doyle & Bottomley, 2006). The accounts diverge most sharply for mismatched pairings—a lease set in a breezy sans-serif, or a friend's note set in a formal serif—where the intrinsic account predicts the typeface's usual effect but the congruence account predicts a penalty.

Why might readers differ? Easily processed, familiar stimuli are judged more favorably, including as more credible and true (Alter & Oppenheimer, 2009; Reber et al., 1998; Song & Schwarz, 2008; Zajonc, 1968). The typographic conventions a reader treats as expected for a given register are learned from the media they have encountered, and cohorts differ in whether consequential messages were first met in print, where serif faces dominated, or on screen, where sans-serif faces prevail. We therefore expected cohorts to differ not necessarily in overall font preference but in how strongly typographic

fit governs their credibility judgments. We treat exposure history as a candidate mechanism rather than a demonstrated one, and included a self-reported index of daily screen reading to test whether recent exposure, rather than cohort itself, carries any difference (Prensky, 2001; Helsper & Eynon, 2010).

Credibility can also be conflated with a reader's sense of what is stylistically appropriate. A reader may know that a serif face is the conventional dress for a contract without that knowledge raising trust. We therefore measured perceived appropriateness alongside credibility, letting convention knowledge and trust come apart. Preregistered predictions were that (H1) credibility would depend on the typeface $\times$ register interaction more than on typeface alone, and (H2) this congruence sensitivity would be stronger in older adults; appropriateness and a reading-ease measure were analyzed as secondary, exploratory outcomes, with no directional prediction for cohort differences in appropriateness.

Method

Participants and design. Two hundred fifty-two U.S. adults were recruited through Prolific; six who failed an embedded attention check were excluded, leaving 246 (123 younger, 18–25 years, $M = 21.4$; 123 older, 58–74 years, $M = 65.1$; all fluent English speakers with normal or corrected-to-normal vision). The study was preregistered. Exemplar family was manipulated between subjects: each participant was randomly assigned to one of three serif/sans-serif families—Times New Roman/Arial, Georgia/Helvetica, or Garamond/Calibri (82 per family)—so that any effect could be attributed to the serif/sans-serif category rather than to one familiar contrast. Typeface (serif vs. sans-serif) and register (institutional vs. interpersonal) were manipulated within subjects. Daily screen-reading time was measured, not manipulated, and treated as a continuous individual-difference covariate.

Materials. Eight short messages were written: four institutional (a residential lease clause and a bank account-security notice, two items each) and four interpersonal (a note from a friend and a message from a coworker, two items each). For each participant, half of the items in each register appeared in the assigned family's serif member and half in its sans-serif member, with item-to-typeface assignment counterbalanced across participants so that every item appeared in both typefaces across the sample while no participant saw the same wording twice. Point sizes were adjusted so that serif and sans-serif exemplars had matched x-heights, reducing raw size differences between conditions. Each message–typeface combination was classified *a priori* as congruent (serif–institutional, sans–interpersonal) or mismatched (serif–interpersonal, sans–institutional) from prevailing print and screen conventions.

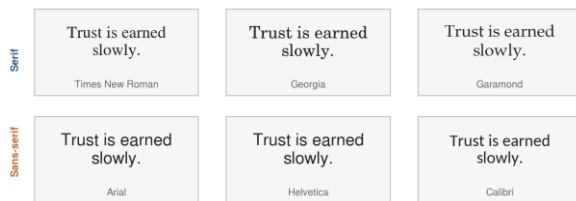


Figure 1. The three serif/sans-serif exemplar families, shown in matched sample text. Participants were randomly assigned to one family; the typeface category, not the individual face, was the variable of interest.

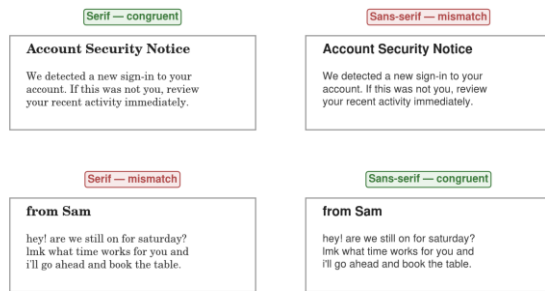


Figure 2. Example stimuli illustrating the typeface × register crossing (Georgia/Helvetica family shown). Identical institutional and interpersonal messages appear in serif and sans-serif, yielding congruent and mismatched pairings.

Measures and analysis. After each message, participants rated its credibility (“How trustworthy is this message?”), then its appropriateness (“How well does the typeface suit this kind of message?”), and finally its reading ease, each on a 1–7 scale; credibility was always rated first to keep the appropriateness question from priming it. Credibility was the primary, preregistered outcome; appropriateness and reading ease were secondary and exploratory. Data were analyzed with linear mixed models including typeface, register, and cohort and their interactions, with random intercepts for participant and item; exemplar family and screen-reading time were entered as preregistered moderators.

Results

Averaged across contexts, serif and sans-serif messages were rated about equally credible: there was *no* reliable overall effect of typeface. That aggregate null is

misleading, however, because the predicted typeface × register interaction was reliable—congruent pairings were rated more credible than mismatched ones (H1). Whether a serif or a sans-serif face raised credibility therefore depended substantially on the message it carried.

This congruence effect was larger in older adults (a typeface × register × cohort interaction; H2), but the cohort difference is not the intuitive one. It is *not* that older adults favored serif and younger adults favored sans-serif. As Table 1 shows, older adults lost trust in both directions when typeface clashed with register—a sans-serif bank notice and a serif message from a friend were each downgraded. Younger adults showed the congruence effect too, but concentrated in interpersonal messages: they penalized a stiff serif on a casual note, yet a sans-serif on an institutional message cost them little credibility. The cohort difference is therefore better described as younger adults tolerating sans-serif in formal contexts, narrowing the range over which mismatch matters.

A dissociation. The secondary appropriateness ratings tell a different story from credibility. Both cohorts rated congruent pairings as far more appropriate than mismatched ones, and this appropriateness gap did *not* differ by cohort (Fig. 3 shows the credibility interaction; Table 1 places the two outcomes side by side). Younger adults recognized that, say, a sans-serif lease violates convention—they judged it less appropriate—yet did not correspondingly lower its credibility. Convention knowledge was shared across cohorts; its translation into a trust judgment was not.

Generalization and null moderations. Exemplar family did not moderate any effect: the pattern held across all three serif/sans-serif families, consistent with an effect of the broad category rather than one familiar typeface pair. Two preregistered checks constrain interpretation. Self-reported screen-reading time did not moderate the credibility effects and did not account for the cohort difference—within each cohort, heavier and lighter screen readers behaved similarly—so recent exposure, at least as indexed here, does not explain why older adults penalize mismatch more. And reading-ease ratings were uniformly high and near-identical across congruent and mismatched conditions, indicating that the congruence effect on credibility was not a by-product of some conditions being harder to read.

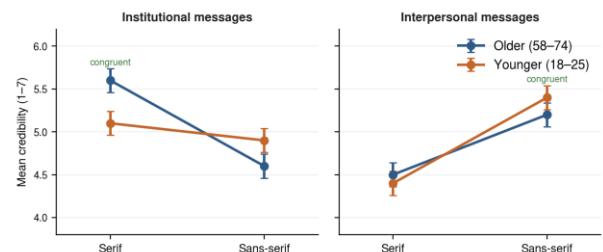


Figure 3. Mean credibility (the primary outcome) by cohort, message register, and typeface, collapsed across exemplar family; congruent pairings are labeled. Older adults lose credibility from a mismatch in either

direction, whereas younger adults tolerate sans-serif on institutional messages. Error bars are ± 1 SE.

Message · typeface (fit)	Older Cred.	Older Appr.	Younger Cred.	Younger Appr.
Institutional · serif (congruent)	5.6	6.0	5.1	5.8
Institutional · sans (mismatch)	4.6	4.3	4.9	4.4
Interpersonal · serif (mismatch)	4.5	3.9	4.4	4.0
Interpersonal · sans (congruent)	5.2	5.7	5.4	5.9

Table 1. Mean credibility and appropriateness ratings (1–7) by cohort, message register, and typeface, collapsed across exemplar family. Congruent rows are shaded. The credibility gap between congruent and mismatched pairings is larger for older adults and, among younger adults, is concentrated in interpersonal messages; the appropriateness gap is similar across cohorts even where credibility gaps differ.

Conclusion

Typeface credibility was shaped more by the fit between a typeface and a message's communicative register than by a general preference for serif or sans-serif fonts. Older adults penalized mismatched typography in both institutional and interpersonal messages, whereas younger adults were more tolerant of sans-serif typefaces in institutional contexts but still penalized serif typefaces in interpersonal messages.

Appropriateness judgments showed that both cohorts recognized similar typographic conventions, even when those conventions influenced their credibility judgments differently. The findings therefore suggest that recognizing a typeface as contextually appropriate and allowing that fit to affect trust are related but distinct processes.

Overall, typefaces do not appear to carry credibility independently of context. Their effect depends on the kind of message they present and on how strongly readers use typographic fit when judging trust.

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